The effect of dietary factors on nitrosoproline levels in human urine.

The effect of dietary components on the levels of nitrosoproline (NPRO) excreted over a 24 h period in the urine was examined in volunteers ingesting known amounts of various food products. The ingestion of nitrite-preserved meats (85-170 g per meal), including canned, rolled or Yunnan ham, cured pork, luncheon meat, and various Chinese and European-style sausages, led to urinary NPRO excretion levels ranging from 2.5 to 78.5 micrograms/24 h, whereas the consumption of non-preserved meat and fish products, including chicken, herring, salmon, shrimp, ground beef (hamburger), pork chops and beef liver, led to relatively low NPRO excretion levels, ranging from 0.0 to 0.8 micrograms/24 h. The urinary NPRO levels of 22 vegetarians and 14 lacto-vegetarians averaged 0.8 and 1.4 micrograms/24 h, respectively. A change from a nitrite-preserved meat diet to a vegetarian diet was accompanied by an approximately six-fold reduction in urinary NPRO levels; however, these remained above control levels for at least 3 days following the dietary change. The relatively high NPRO levels following the ingestion of nitrite-preserved meats could not be reduced by nitrite-trapping chemicals, including ascorbic acid, ferulic acid, caffeic acid, or phenolic-containing mixtures such as coffee and tea, which were effective in suppressing endogenous NPRO formation following the intake of nitrate and proline. The high urinary NPRO levels after ingestion of preserved meat products appear to be due to the consumption of preformed NPRO . An understanding of the relative contribution of preformed and endogenously formed nitrosamines appears to be essential when designing dietary intervention programmes.